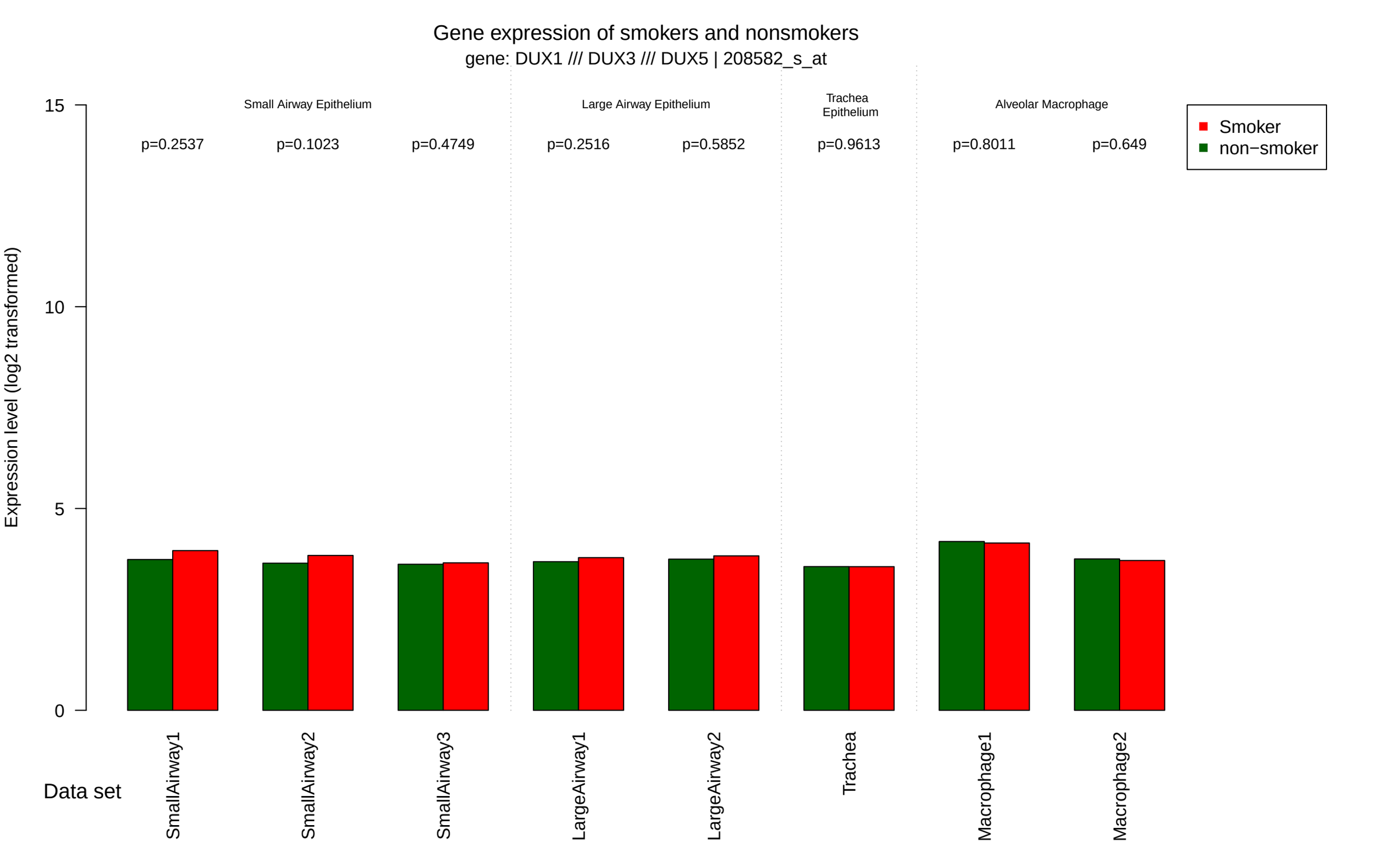
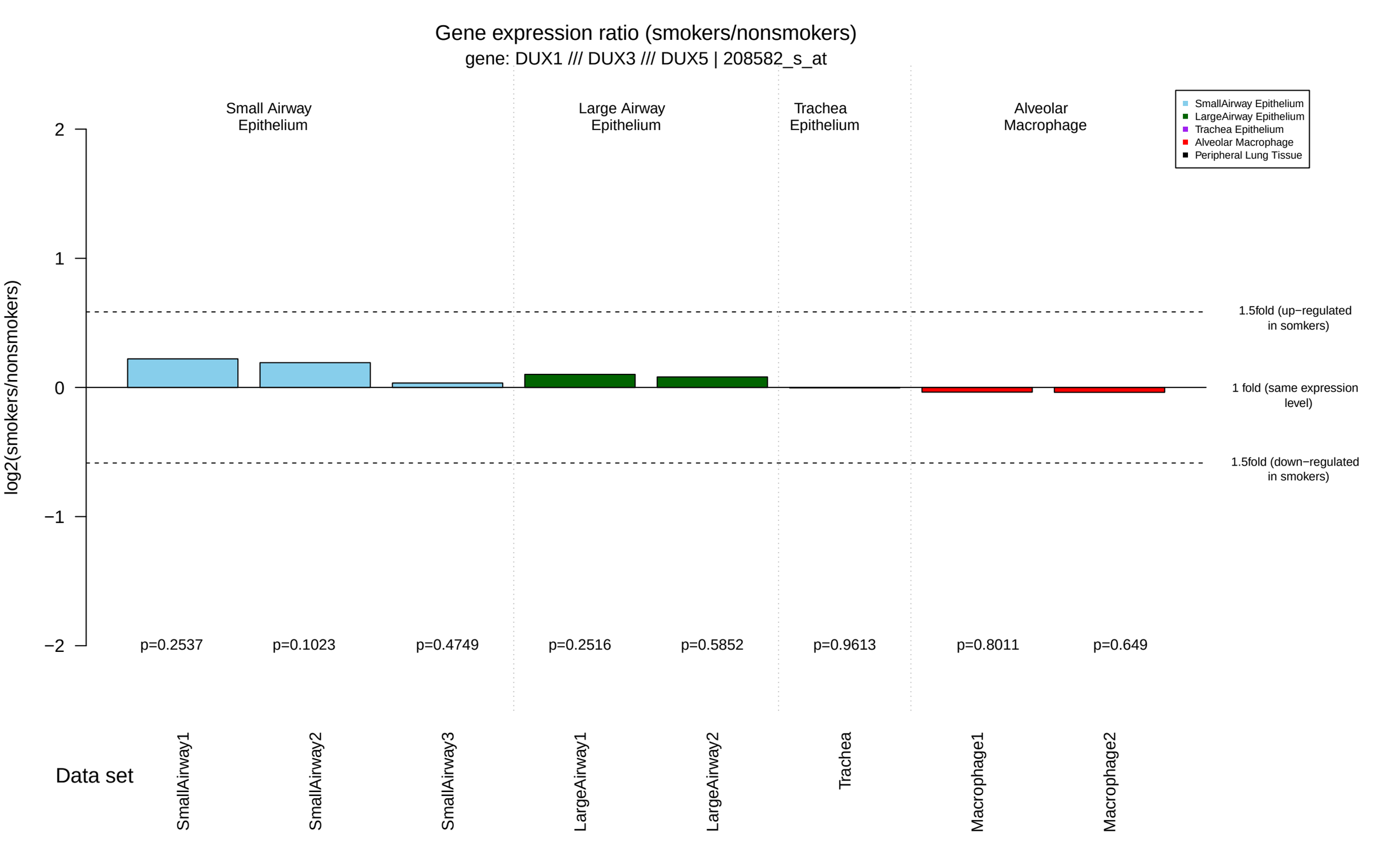
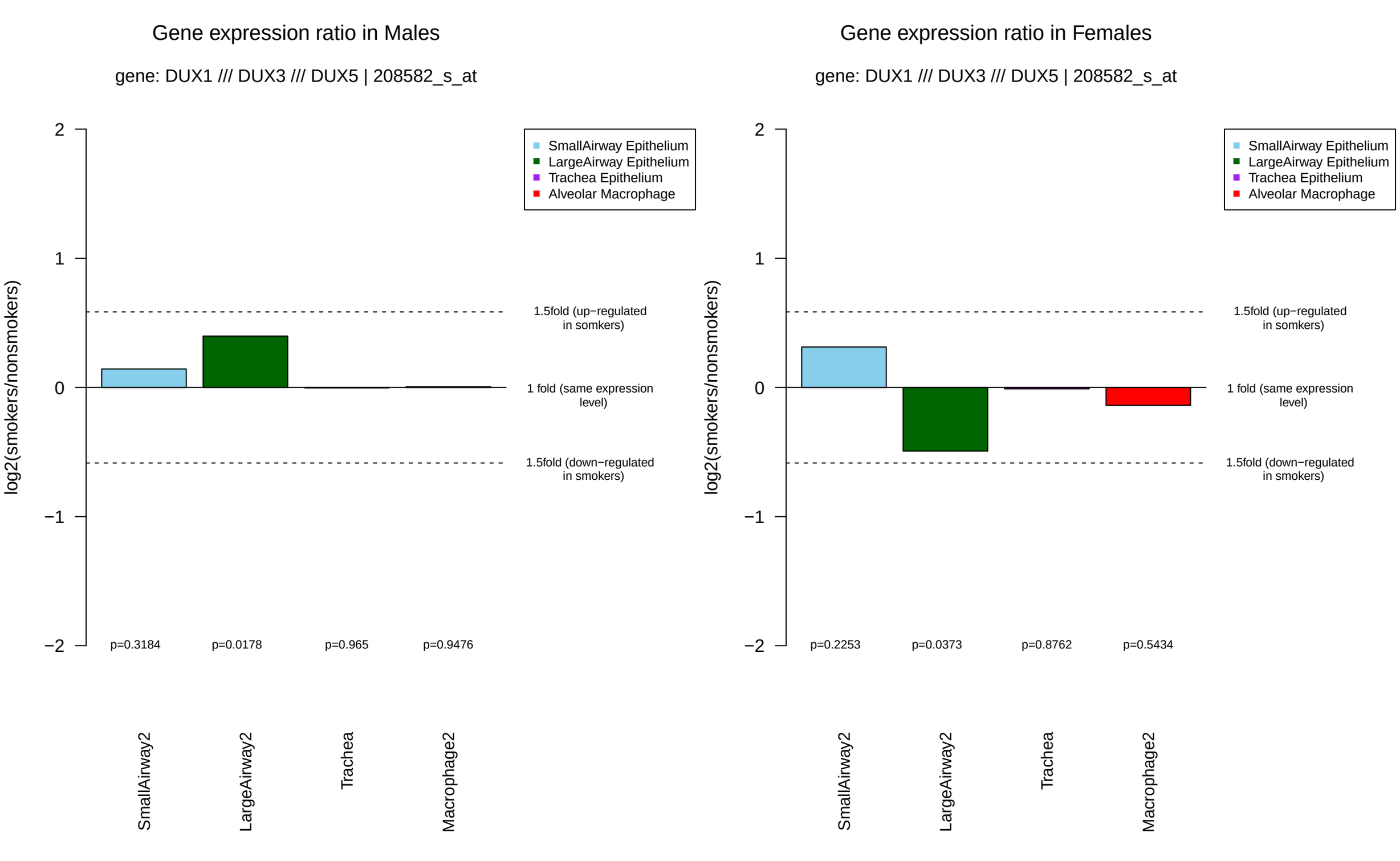
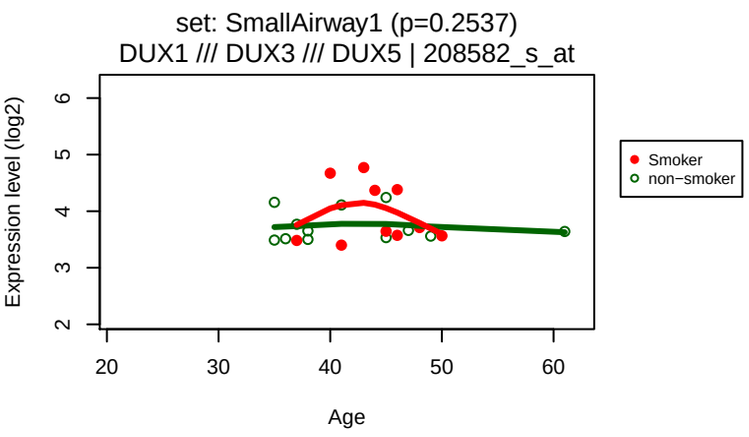
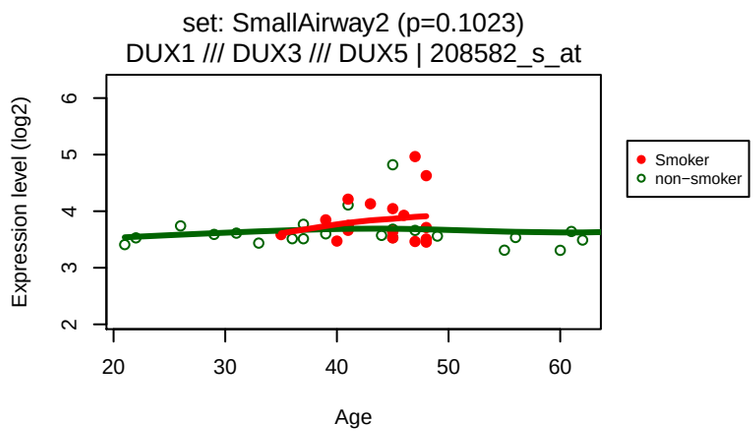
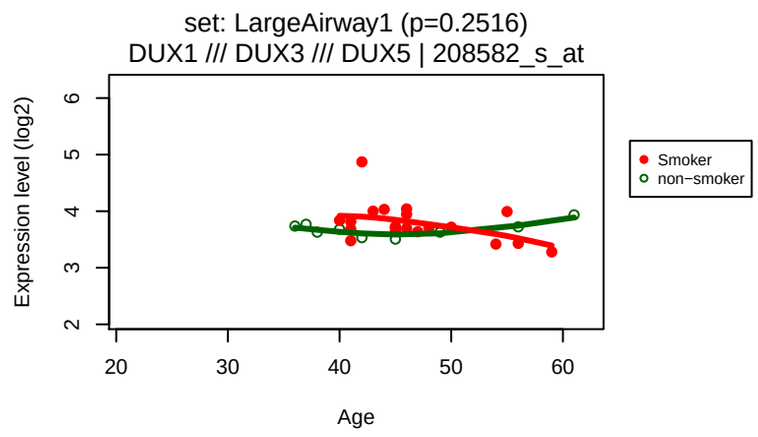

Gene expression of smokers and nonsmokers
gene: DUX1 /// DUX3 /// DUX5 | 208582_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2537
p=0.1023
p=0.4749
p=0.2516
p=0.5852
p=0.9613
p=0.8011
p=0.649
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

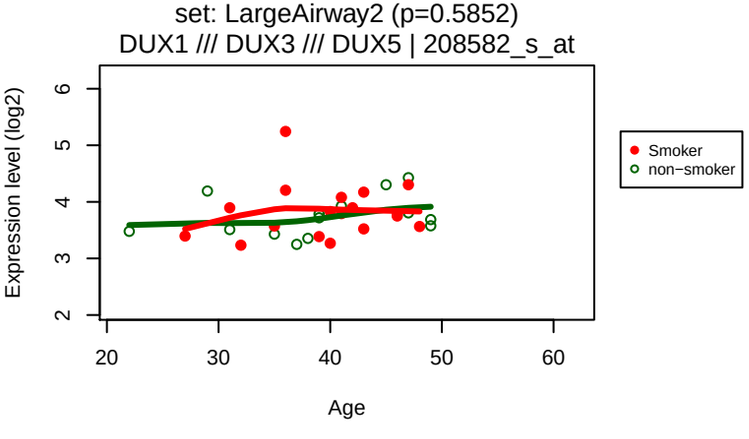

Gene expression ratio (smokers/nonsmokers)
gene: DUX1 /// DUX3 /// DUX5 | 208582_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2537
p=0.1023
p=0.4749
p=0.2516
p=0.5852
p=0.9613
p=0.8011
p=0.649
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

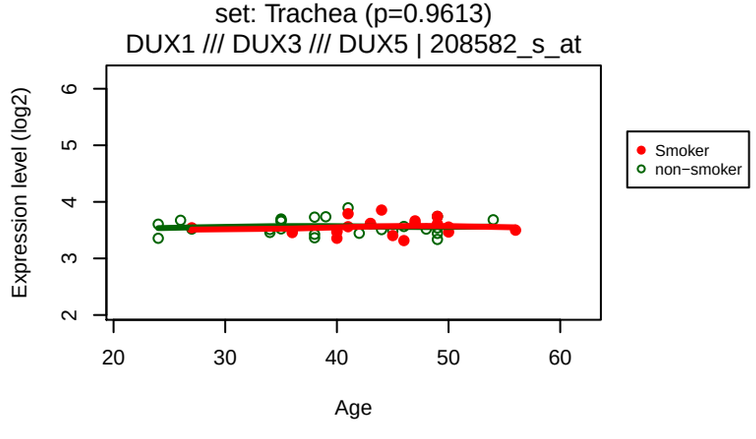

Gene expression ratio in Males
gene: DUX1 /// DUX3 /// DUX5 | 208582_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3184
p=0.0178
p=0.965
p=0.9476
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DUX1 /// DUX3 /// DUX5 | 208582_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2253
p=0.0373
p=0.8762
p=0.5434
SmallAirway2
LargeAirway2
Trachea
Macrophage2

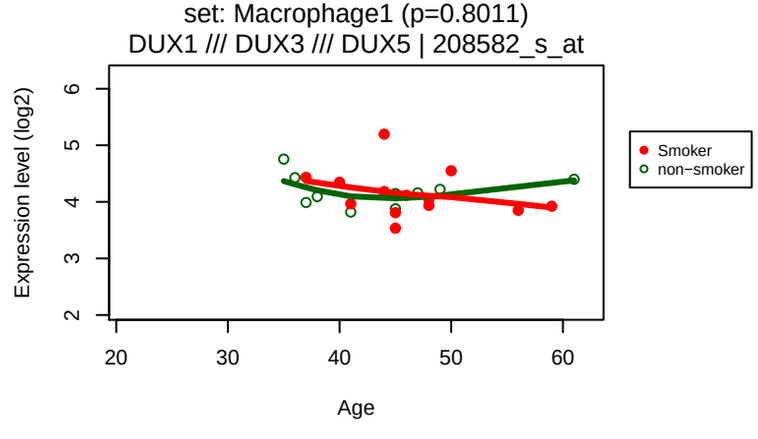

set: SmallAirway1 (p=0.2537)
DUX1 /// DUX3 /// DUX5 | 208582_s_at
Expression level (log2)
Age
Smoker
non−smoker

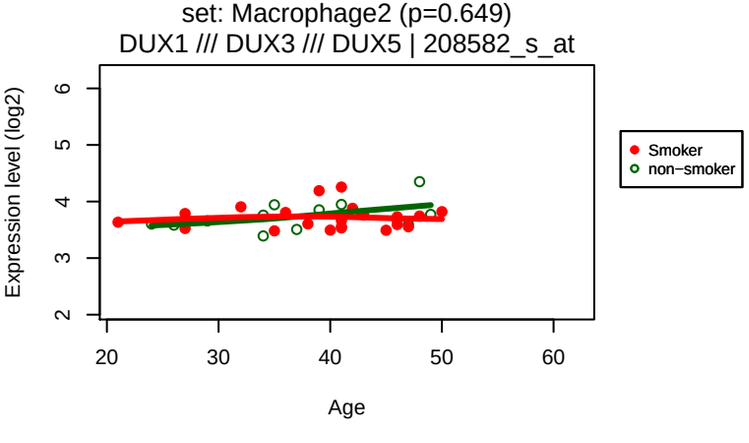

set: SmallAirway2 (p=0.1023)
DUX1 /// DUX3 /// DUX5 | 208582_s_at
Expression level (log2)
Age
Smoker
non−smoker

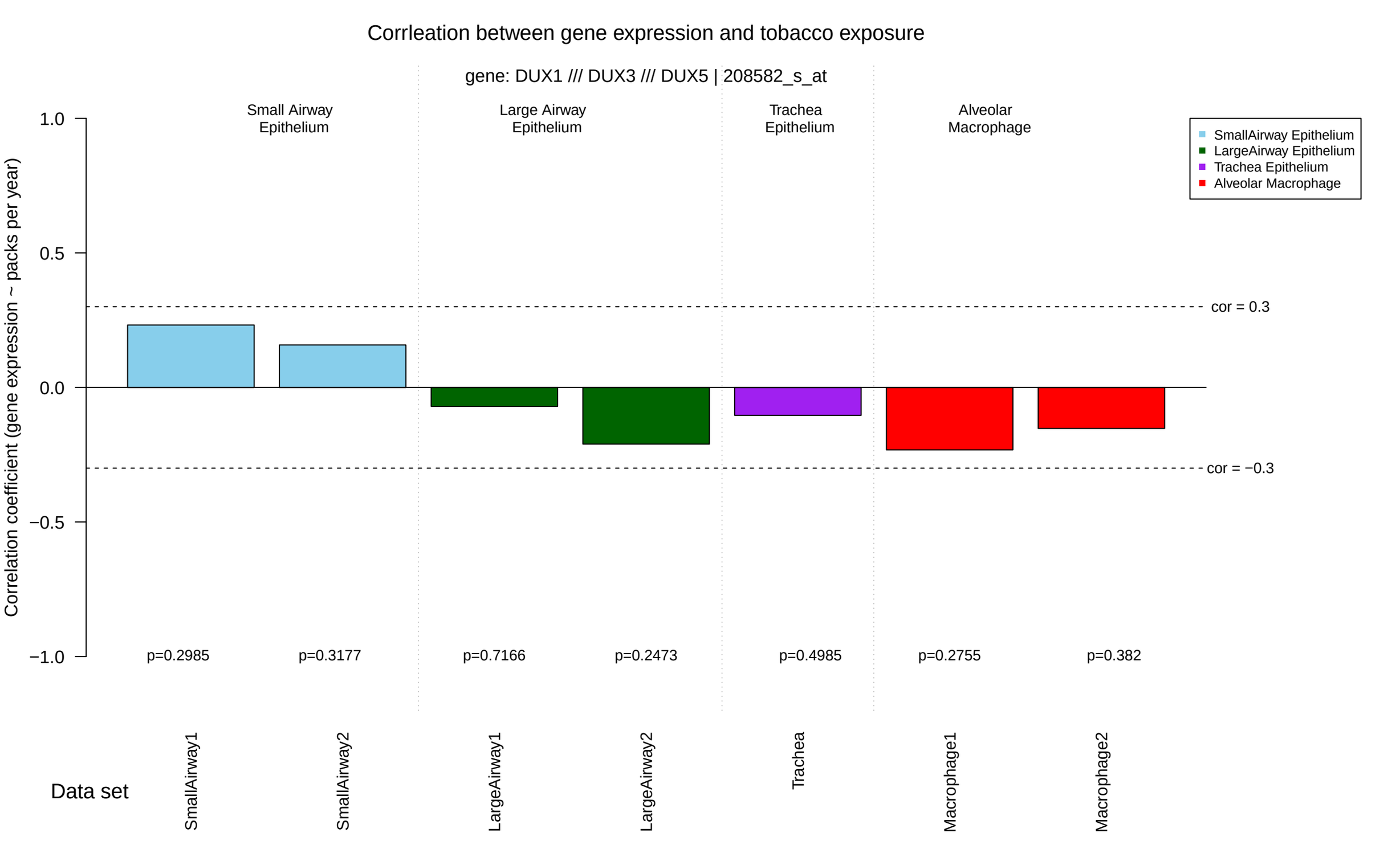

set: LargeAirway1 (p=0.2516)
DUX1 /// DUX3 /// DUX5 | 208582_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5852)
DUX1 /// DUX3 /// DUX5 | 208582_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9613)
DUX1 /// DUX3 /// DUX5 | 208582_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8011)
DUX1 /// DUX3 /// DUX5 | 208582_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.649)
DUX1 /// DUX3 /// DUX5 | 208582_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DUX1 /// DUX3 /// DUX5 | 208582_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2985
p=0.3177
p=0.7166
p=0.2473
p=0.4985
p=0.2755
p=0.382
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2